

FLOOD PREDICTION SYSTEM USING LSTM NEURAL NETWORKS AND RAINFALL DATA

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ABSTRACT

This study implements an LSTM-based flood prediction model using 10 years of rainfall and water level data from Cavite Province, achieving 92.3% prediction accuracy 24 hours ahead of flood events.

INTRODUCTION

Cavite Province experiences regular monsoon-driven flooding causing property damage and casualties. Current PAGASA forecasts lack local granularity. This study builds an LSTM neural network for Imus River flood prediction using historical rainfall and water level telemetry data, targeting 24-hour advance warning with high accuracy for local DRRMO use.

METHODOLOGY

Ten years (2013-2022) of 6-hourly rainfall data (mm) and river water level (m) from 3 PAGASA stations were used. Data preprocessing: linear interpolation for missing values, MinMax normalization. LSTM architecture: 2 stacked layers (128 and 64 units), dropout=0.2, lookback window=48 timesteps (12 days). Output: binary flood/no-flood classification for $t+24h$. Train/validation/test: 2013-2020/2021/2022. Evaluation: accuracy, precision, recall, F1, AUC-ROC.

RESULTS

LSTM achieved 92.3% accuracy, precision 88.7%, recall 94.1%, F1 91.3%, AUC-ROC 0.96. Compared to ARIMA (accuracy 79%) and persistence model (72%), LSTM demonstrated clear superiority. Average prediction lead time was 26.4 hours. The model predicted 14/15 flood events in 2022, missing one flash flood event caused by an isolated convective storm.

DISCUSSION

LSTM's ability to capture long-term temporal dependencies is critical for flood forecasting in a monsoon climate. The missed prediction was associated with a localized convective storm pattern not well-represented in training data. Future work should incorporate satellite rainfall estimates and real-time weather model outputs to handle such events.